

Modelo de optimización para la integración de operaciones en curso y de emergencia

Antonio Sevilla Sastre



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01

Planteamiento del problema





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Integrating supply chains for emergencies and ongoing operations in UNHCR



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<https://www.sciencedirect.com/science/article/pii/S0272696316300444>





- **Operaciones sostenidas, en curso**



- **Operaciones de emergencia**



Integrar

- Operaciones sostenidas, en curso
- Operaciones de emergencia

Diferentes demandas



Diferentes demandas



En curso

Diferentes demandas



En curso

- Homogénea
- Predecible
- Aplazable

Diferentes demandas



En curso

- Homogénea
- Predecible
- Aplazable



Emergencia

Diferentes demandas



En curso

- Homogénea
- Predecible
- Aplazable



Emergencia

- Heterogénea
- Incertidumbre
- Urgente

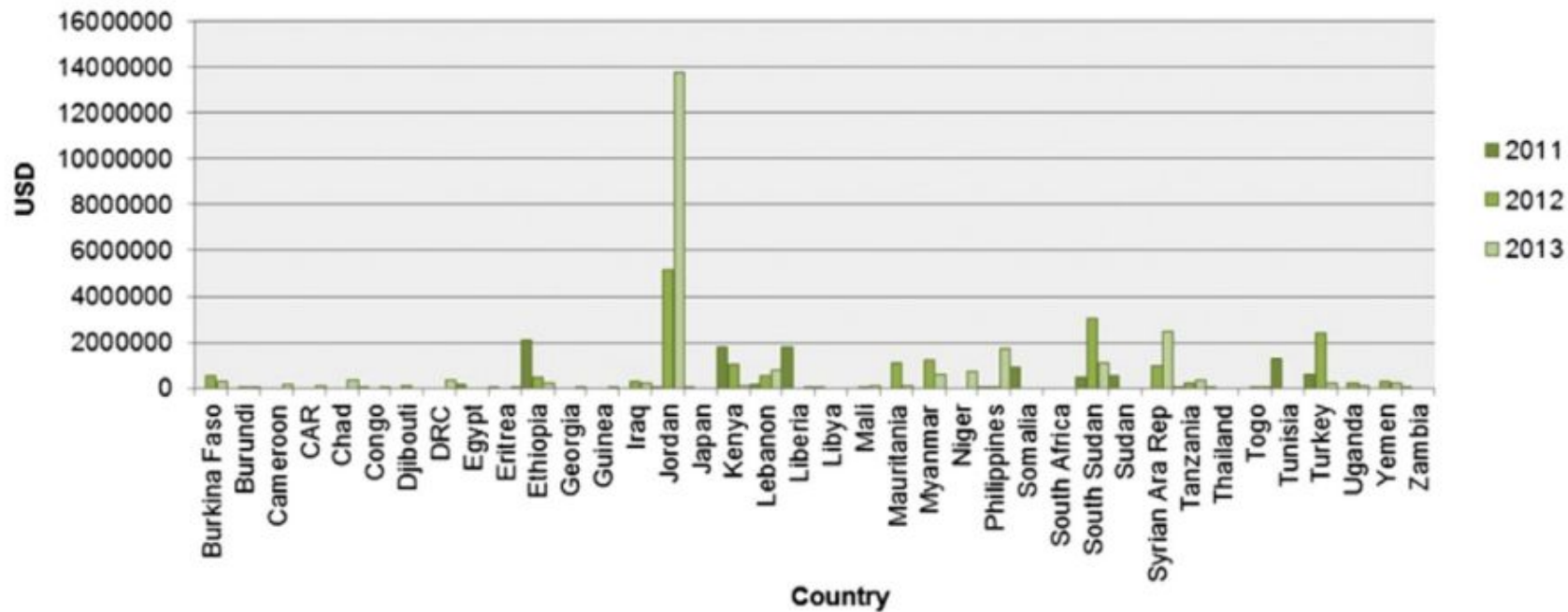


Fig. 3. Variability in ER demand over 2011–2013 (three-month average) per country.

Desplazados forzados en el mundo por país de acogida





Fig. 2. Warehouses in UNHCR's network (June 2015).

¡Stock preposicionado
para **ambos** tipos
de operaciones!

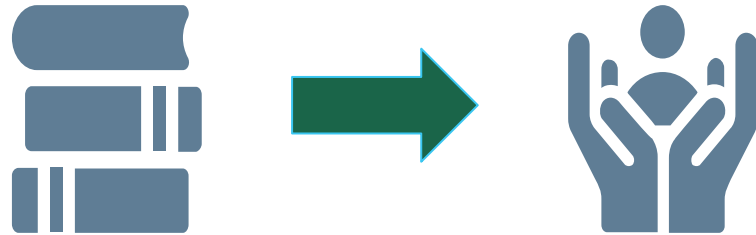


Fig. 2. Warehouses in UNHCR's network (June 2015).

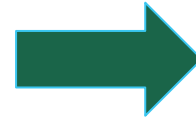
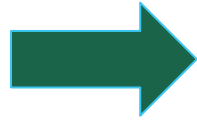
Diferentes flujos de transporte



Diferentes flujos de transporte



Diferentes flujos de transporte



Diferentes flujos de transporte



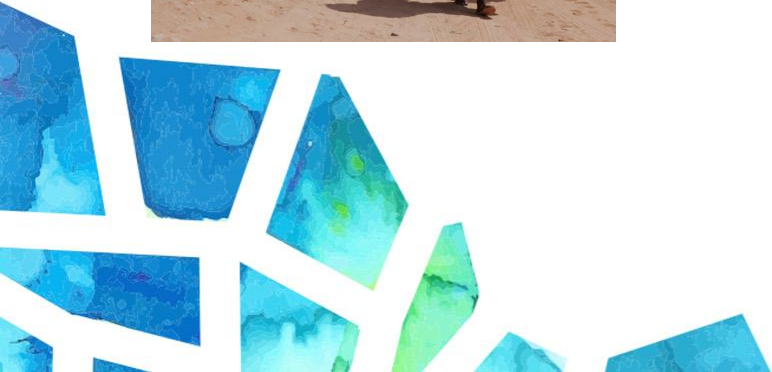
Diferentes costes integrados



Diferentes costes integrados



Accesibilidad



Diferentes costes integrados



Accesibilidad



Seguridad



Diferentes costes integrados



Accesibilidad



Seguridad



Co-localización









02

Explicación del modelo



Conjuntos

S	<i>Set of suppliers</i>
G	<i>Set of potential global warehouses</i>
A	<i>Set of all demand points including ongoing and emergency</i>
K	<i>Set of emergency response scenarios</i>

Parámetros

f_g	<i>Fixed cost of opening a potential global warehouses g in USD</i>
v_g	<i>Variable cost of opening a potential global warehouses g in USD</i>
C_{gj}^{ER}	<i>Transportation cost from global warehouse g to demand point j using express air shipment in USD</i>
CL_{gj}^{ER}	<i>Transportation cost from global warehouse g to demand point j using express land shipment in USD</i>
C_{sj}^{oo}	<i>Transportation cost from supplier s to demand point j using normal shipment in USD</i>
C_{gj}^{oo}	<i>Transportation cost from global warehouse g to demand point j using normal shipment in USD</i>
C_{sg}	<i>Transportation cost from supplier s to global warehouse g using normal shipment in USD</i>

Parámetros

l_{gj}^e	Lead time from global warehouse g to demand point j using express air shipment in days
lL_{gj}^e	Lead time from global warehouse g to demand point j using express land shipment in days
l_{gj}^n	Lead time from global warehouse g to demand point j using normal shipment in days
l_{sj}^n	Lead time from Supplier s to demand point j using normal shipment in days
d_j^{oo}	Demand for ongoing operations at location j in USD
$d_j^{ER,k}$	Demand for emergency response at location j under scenario k in USD
lim	Equal to 430,000 USD, the dollar value of ten TEU's
Q_g	A large number, or capacity of g 'th global warehouse in TEU

Variables

I_g	<i>Required number of containers to be stocked in global warehouse g in TEU</i>
z_{sg}	<i>Number of containers to send from supplier s to global warehouse g with normal shipment in TEU</i>
$x_{sj}^{oo,k}$	<i>Percentage of ongoing demand to send from supplier s to demand point j with normal shipment in the k'th scenario</i>
$x_{gj}^{oo,k}$	<i>Percentage of ongoing demand to send from global warehouse g to demand point j with normal shipment in the k'th scenario</i>
$x_{gj}^{ER,k}$	<i>Percentage of emergency demand to send from global warehouse g to demand point j with express air shipment in the k'th scenario</i>
$ex_{gj}^{ER,k}$	<i>Excess percentage of emergency demand to send from global warehouse g to demand point j with express land shipment (after sending 10 TEU's by express air) in the k'th scenario</i>

Variables

y_g	1, if global warehouse g is opened 0, otherwise
bz_{sg}	1, if containers are sent from suppliers s to global warehouse g 0, otherwise
$bx_{sj}^{oo,k}$	1, if containers are sent from supplier s to demand point j by normal shipment 0, otherwise
$bx_{gj}^{oo,k}$	1, if containers are sent from global warehouse g to demand point j by normal shipment 0, otherwise
$bx_{gj}^{ER,k}$	1, if containers are sent from global warehouse g to demand point j by express air shipment 0, otherwise
$bex_{gj}^{ER,k}$	1, if excess containers are sent from global warehouse g to demand point j by express land shipment 0, otherwise

Funciones objetivo

$$\min \sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right. \\ \left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - lim) \right) \right)$$

$$\min \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\sum_{g \in G} l_{gj}^n b x_{gj}^{oo,k} + \sum_{s \in S} l_{sj}^n b x_{sj}^{oo,k} + \sum_{g \in G} l_{gj}^e b x_{gj}^{ER,k} + \sum_{g \in G} l L_{gj}^e b e x_{gj}^{ER,k} \right)$$

Restricciones

$$\sum_{g \in G} (x_{gj}^{oo,k} d_j^{oo}) + \sum_{s \in S} (x_{sj}^{oo,k} d_j^{oo}) = d_j^{oo},$$

$$\sum_{g \in G} (x_{gj}^{ER,k} d_j^{ER,k} + ex_{gj}^{ER,k} d_j^{ER,k}) = d_j^{ER,k},$$

Restricciones

$$\sum_{g \in G} (x_{gj}^{ER,k} d_j^{ER,k}) = (1 - \gamma_j^k) \lim + d_j^{ER,k} \gamma_j^k,$$

$$\sum_{g \in G} (ex_{gj}^{ER,k} d_j^{ER,k}) = (1 - \gamma_j^k) (d_j^{ER,k} - \lim),$$

Restricciones

$$I_g \leq Q_g y_g,$$

$$\sum_{s \in S} z_{sg} = I_g,$$

$$\sum_{j \in A} (x_{gj}^{oo,k} d_j^{oo} + x_{gj}^{ER,k} d_j^{ER,k} + ex_{gj}^{ER,k} d_j^{ER,k}) \leq 43000 I_g,$$

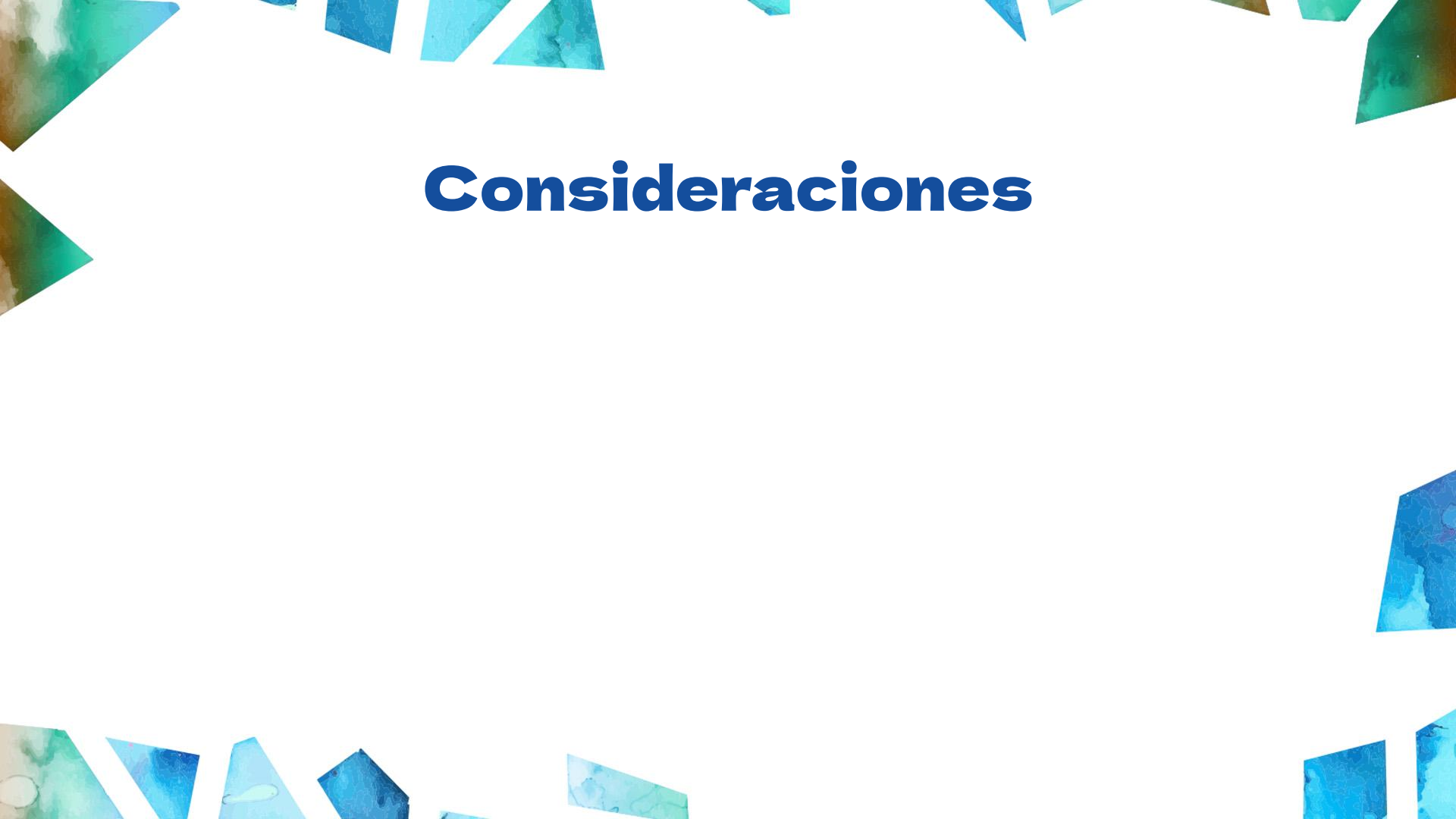
Restricciones

$$bx_{gj}^{oo,k} \geq x_{gj}^{oo,k},$$

$$bx_{sj}^{oo,k} \geq x_{sj}^{oo,k},$$

$$bx_{gj}^{ER,k} \geq x_{gj}^{ER,k},$$

$$bex_{gj}^{ER,k} \geq ex_{gj}^{ER,k},$$



Consideraciones

Consideraciones

- Facility Location Problem (FLP)

Consideraciones

- Facility Location Problem (FLP)
- Demanda inflexible

Consideraciones

- Facility Location Problem (FLP)
- Demanda inflexible
- Sin restricciones de carga

Consideraciones

- Facility Location Problem (FLP)
- Demanda inflexible
- Sin restricciones de carga
- MIP

03

Técnicas empleadas



Programación estocástica

Programación estocástica



Programación estocástica

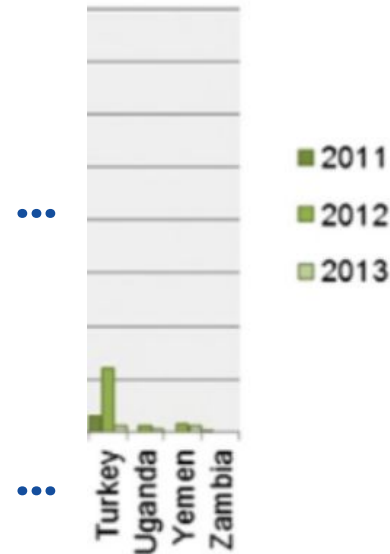
Integrar distintos escenarios



Programación estocástica



Integrar distintos escenarios



Programación estocástica

$$\min \sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right. \\ \left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - lim) \right) \right)$$

$$\min \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\sum_{g \in G} l_{gj}^n b x_{gj}^{oo,k} + \sum_{s \in S} l_{sj}^n b x_{sj}^{oo,k} + \sum_{g \in G} l_{gj}^e b x_{gj}^{ER,k} + \sum_{g \in G} l L_{gj}^e b e x_{gj}^{ER,k} \right)$$

Programación estocástica

$$\min \sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - lim) \right) \right)$$

$$\min \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\sum_{g \in G} l_{gj}^n b x_{gj}^{oo,k} + \sum_{s \in S} l_{sj}^n b x_{sj}^{oo,k} + \sum_{g \in G} l_{gj}^e b x_{gj}^{ER,k} + \sum_{g \in G} l L_{gj}^e b e x_{gj}^{ER,k} \right)$$



Programación multicriterio

Programación multicriterio

Tener en cuenta distintos criterios



Programación multicriterio

Tener en cuenta distintos criterios



Programación multicriterio

Tener en cuenta distintos criterios
frecuentemente enfrentados





Programación por fases

Programación por fases

1. Minimizar costes
2. Minimizar tiempos
sin empeorar costes



Programación por fases

1. Minimizar costes



2. Minimizar tiempos
sin empeorar costes



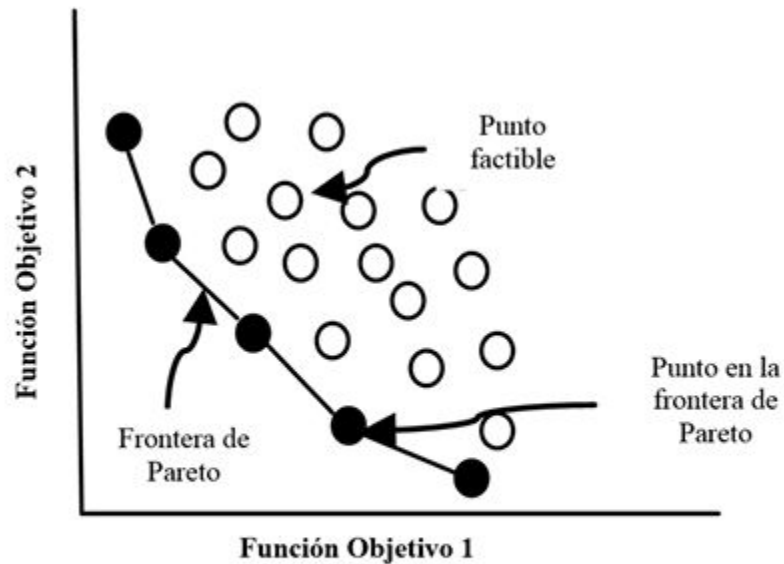
¡El orden tiene implicaciones éticas!

Programación por fases

1. $\min F1$ s.a. $R1 \rightarrow F1^*$
2. $\min F2$ s.a. $R1, R2, y$
 $F1 \leq F1^*$

Frontera de Pareto

Frontera de Pareto



Soluciones Dominadas



Soluciones No Dominadas

Frontera de Pareto

1. $\min F1$ s.a. $R1 \rightarrow F1^*$
2. $\min F2$ s.a. $R1, R2, y$
 $F1 \leq F1^* \cdot (1+\beta)$

Frontera de Pareto

$$\sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right.$$

$$\left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - lim) \right) \right) \leq TC^* (1 + \beta)$$

Frontera de Pareto

$$\sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right.$$

$$\left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) \lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - \lim) \right) \right) \leq TC^*(1 + \beta)$$

Frontera de Pareto

$$\sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right.$$

$$\left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) \lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - \lim) \right) \right) \leq TC^*(1 + \beta)$$

3.5

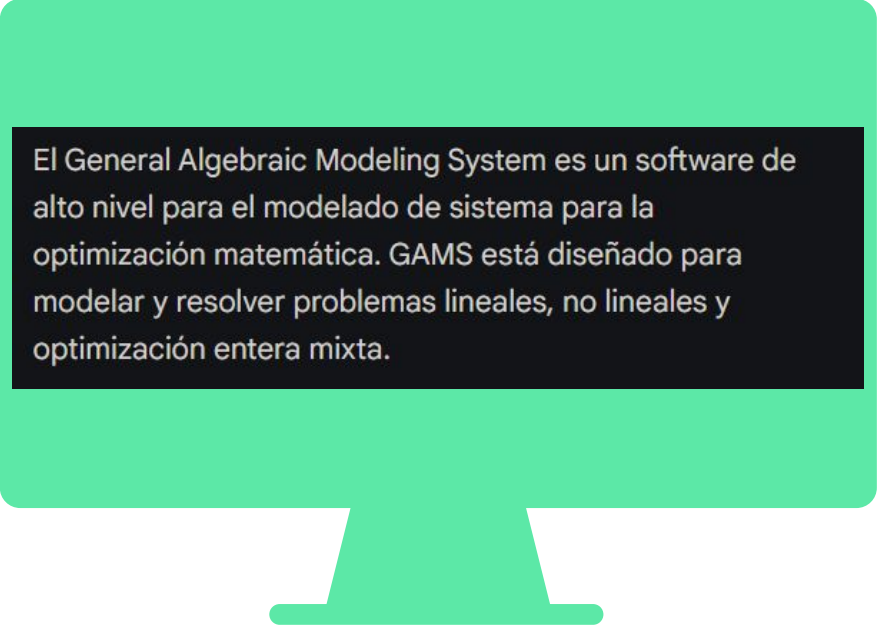
Implementación del modelo en GAMS



Modelo en GAMS



Modelo en GAMS



El General Algebraic Modeling System es un software de alto nivel para el modelado de sistema para la optimización matemática. GAMS está diseñado para modelar y resolver problemas lineales, no lineales y optimización entera mixta.

Modelo en GAMS

```
F1.. F_OBJ_1 =E= sum(g, f(g)*y(g)) + sum((g,s), C(s,g)*z(s,g)) + sum(g,
    sum((k,j), alpha(k)*(
        (sum(g,C_006J(g,j)*x_006J(g,j,k)) + sum(s, C_00SJ(s,
        sum(g, C_ER(g,j)*C1(g,j,k) +
        (CL_ER(g,j)*(d_ER(j,k)-lim)*C2(g,j,k))
    )));

F2.. F_OBJ_2 =E= sum((j,k), alpha(k)*(sum(g, l_n6J(g,j)*bx_006J(g,j,k))
eq_sat_dem_00(j,k) .. sum(g, x_006J(g,j,k)*d_00(j)) + sum(s, x_00SJ(s,j,k)
eq_sat_dem_ER(j,k) .. sum(g, x_ER(g,j,k)*d_ER(j,k) + ex_ER(g,j,k)*d_ER(j,
eq_diez_TEU_1(j,k) .. sum(g, x_ER(g,j,k)*d_ER(j,k)) =E= (1-gamma_(j,k))*L
eq_diez_TEU_2(j,k) .. sum(g, ex_ER(g,j,k)*d_ER(j,k)) =E= (1-gamma_(j,k))*L
eq_req_apertura(g) .. I(g) =L= Q(g)*y(g);
```



<https://github.com/asevilla-sastre/GAMS-FLP-ACNUR>

Modelo en GAMS

$$\begin{aligned} \min \quad & \sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right. \\ & \left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k} ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k} (1 - \gamma_j^k) (d_j^{ER,k} - lim))) \right) \right) \end{aligned}$$

The entire model is not presented for the sake of brevity.

Modelo en GAMS

```
F1.. F_OBJ_1 =E= sum(g, f(g)*y(g)) + sum((g,s), C(s,g)*z(s,g)) + sum(g, v(g)*I(g)) +
    sum((k,j), alpha(k)*(
        (sum(g,C_00GJ(g,j)*x_00GJ(g,j,k)) + sum(s, C_00SJ(s,j)*x_00SJ(s,j,k)))*d_00(j) +
        sum(g, C_ER(g,j)*C1(g,j,k) +
            (CL_ER(g,j)*(d_ER(j,k)-lim)*C2(g,j,k))
        )));

eq_C1_1_d_lower_lim(g,j,k)$(d_ER(j,k)≤lim) .. x_ER(g,j,k)*d_ER(j,k) =L= C1(g,j,k);
eq_C1_2_d_lower_lim(g,j,k)$(d_ER(j,k)≤lim) .. C1(g,j,k) =L= x_ER(g,j,k)*d_ER(j,k) + 1000000000000*(1-gamma_(j,k));
eq_C1_3_d_lower_lim(g,j,k)$(d_ER(j,k)≤lim) .. C1(g,j,k) =L= x_ER(g,j,k)*lim;
eq_C1_4_d_lower_lim(g,j,k)$(d_ER(j,k)≤lim) .. x_ER(g,j,k)*lim =L= C1(g,j,k) + 1000000000000*gamma_(j,k);

eq_C1_1_d_greater_lim(g,j,k)$(d_ER(j,k)>lim) .. x_ER(g,j,k)*lim =L= C1(g,j,k);
eq_C1_2_d_greater_lim(g,j,k)$(d_ER(j,k)>lim) .. C1(g,j,k) =L= x_ER(g,j,k)*lim + 1000000000000*gamma_(j,k);
eq_C1_3_d_greater_lim(g,j,k)$(d_ER(j,k)>lim) .. C1(g,j,k) =L= x_ER(g,j,k)*d_ER(j,k);
eq_C1_4_d_greater_lim(g,j,k)$(d_ER(j,k)>lim) .. x_ER(g,j,k)*d_ER(j,k) =L= C1(g,j,k) + 1000000000000*(1-gamma_(j,k));

eq_C2_1(g,j,k) .. C2(g,j,k) =L= 1000000000000*(1-gamma_(j,k));
eq_C2_2(g,j,k) .. C2(g,j,k) =L= ex_ER(g,j,k);
eq_C2_3(g,j,k) .. ex_ER(g,j,k) =L= C2(g,j,k) + 1000000000000*gamma_(j,k);
```

Modelo en GAMS

```
*** Status: Terminated due to a licensing error
***      Problem size limitations have been exceeded
***      Inspect listing file for more information
```

Modelo en GAMS

```
Model modelo_1 /all/;
```

```
Model modelo_2 /all/;
```

1. min F1 s.a. R1 $\rightarrow$ F1*

2. min F2 s.a. R1, R2, y
 $F1 \leq F1^*$



04

Caso de estudio propuesto







MEDECINS SANS FRONTIERES

En 2023, el presupuesto del ACNUR fue de 10.929 millones de dólares, mientras que los gastos de Médicos Sin Fronteras (MSF) superaron los 281 millones de euros. [🔗](#)



<https://www.msf.es/msf-en-el-mundo>



Si somos capaces de responder a las emergencias en un máximo de 72 horas, al tiempo que mantenemos nuestros proyectos regulares en más de 70 países, es gracias a la profesionalización de nuestra logística y cadena de aprovisionamiento.

Es la gran desconocida y, sin embargo, es la que garantiza que nuestros equipos siempre cuenten, allí donde vayan, con los materiales necesarios para prestar atención médico-humanitaria de forma **rápida, eficaz y adaptada**.

Esta red logística comienza en nuestros **centros internacionales de aprovisionamiento** de Bruselas (Bélgica), Burdeos (Francia) y Ámsterdam (Países Bajos), que abastecen a toda nuestra organización. Además, tenemos un almacén de artículos de primera necesidad y material de cobijo en Dubai (Emiratos Árabes).

<https://www.msf.es/quienes-somos/como-enviamos-ayuda>

Centros logísticos MSF

 **Ámsterdam**

 **Bruselas**

 **Burdeos**

 **Dubái**



Centros logísticos MSF

 **Marruecos**

 **Ghana**

 **Sudáfrica**

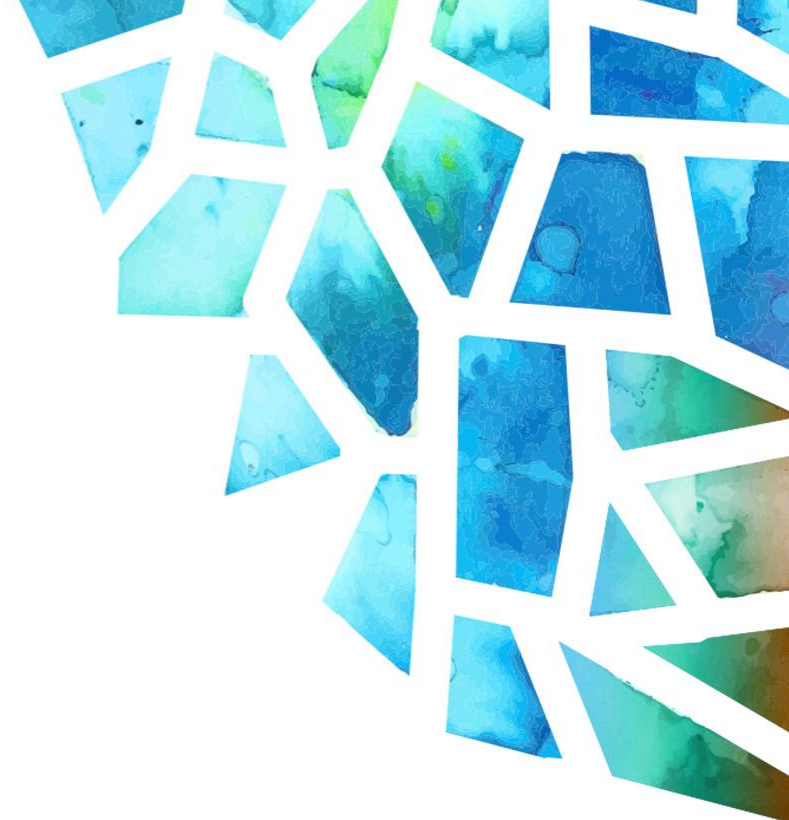




Nos vamos a



G A M S



Frontera de Pareto

$$\sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right.$$

$$\left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - lim) \right) \right) \leq TC^*(1 + \beta)$$

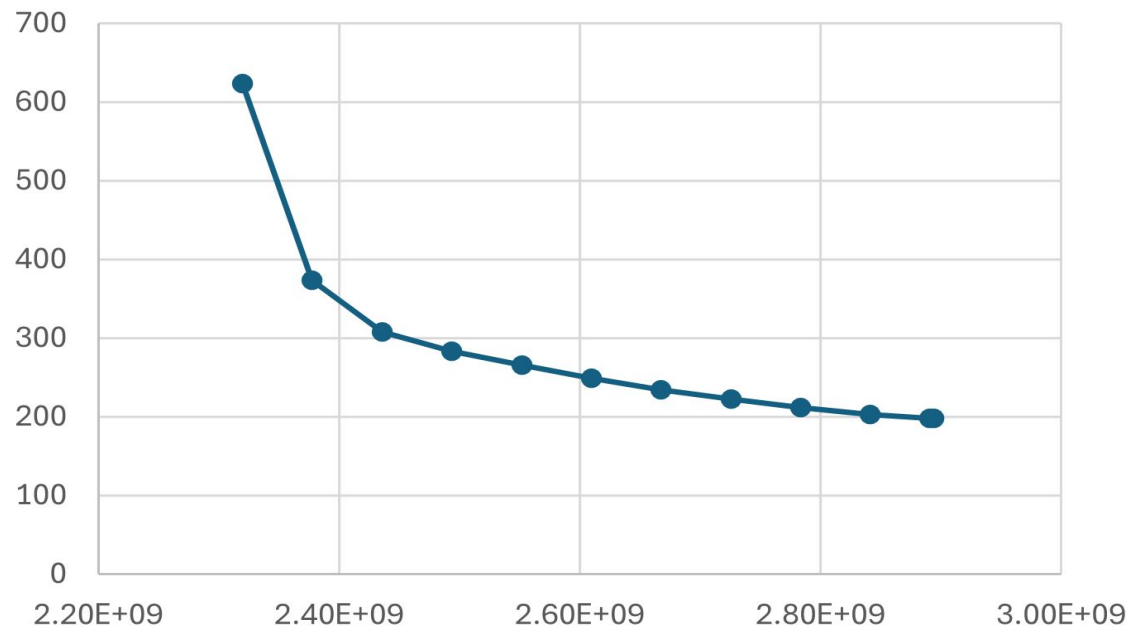
Frontera de Pareto

$$\sum_{g \in G} f_g y_g + \sum_{g \in G} \sum_{s \in S} C_{sg} z_{sg} + \sum_{g \in G} v_g I_g + \sum_{k \in K} \sum_{j \in A} \alpha_k \left(\left(\sum_{g \in G} C_{gj}^{oo} x_{gj}^{oo,k} + \sum_{s \in S} C_{sj}^{oo} x_{sj}^{oo,k} \right) d_j^{oo} + \right. \\ \left. \sum_{g \in G} \left((C_{gj}^{ER} x_{gj}^{ER,k}) ((1 - \gamma_j^k) lim + d_j^{ER,k} \gamma_j^k) + (CL_{gj}^{ER} ex_{gj}^{ER,k}) (1 - \gamma_j^k) (d_j^{ER,k} - lim) \right) \right) \leq TC^* (1 + \beta)$$

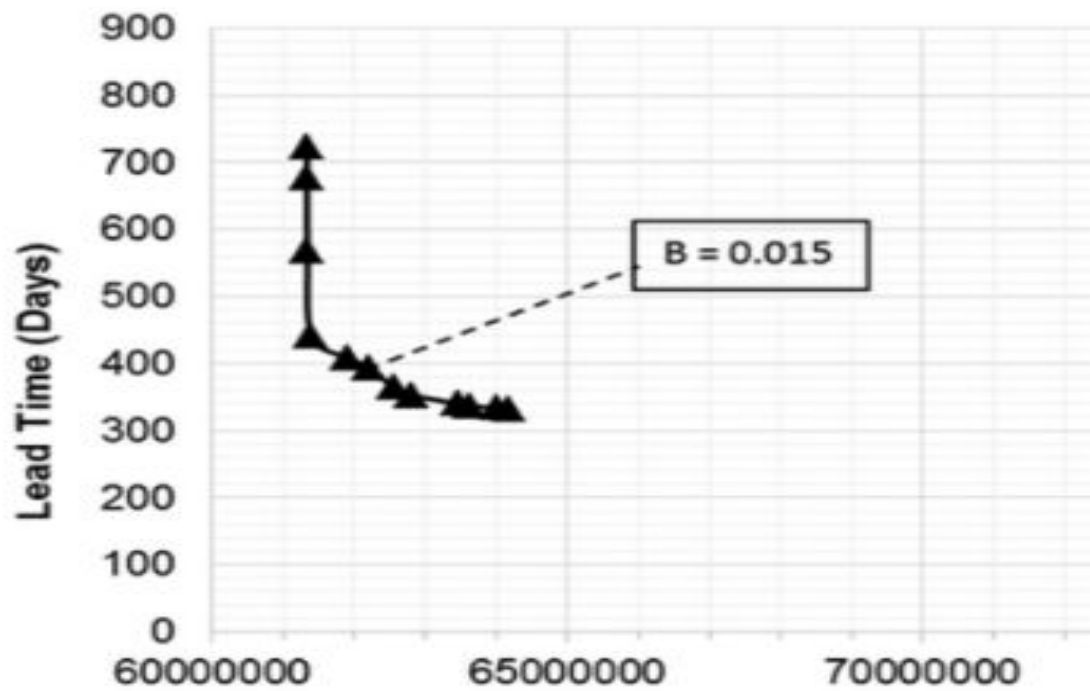
¡Podría hacerse variando tiempos
y optimizando costes!

Tiempo de entrega

Frontera de Pareto



Coste



Presupuesto



Liquidiez

<https://www.sciencedirect.com/science/article/pii/S0272696316300444>

Paper en el que se basa esta presentación

<https://github.com/asevillasastre/GAMS-FLP-ACNUR>

El proyecto en GitHub

<https://www.msf.es/quienes-somos>

Web de Médecins Sans Frontières



Gracias

por vuestra atención

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